Study Script: Formative Study

Black: script; Blue: comments (not read out); Red: actions

Read the Consent

Before we start, I will read the consent form to you.

You are invited to participate in a research study to understand the challenges and needs of low-vision people in daily activities. The purpose of this study is to characterize the needs of people with low vision in activities of daily living, such as what are the important objects they need to perceive, what key components of objects are more important, and the priorities among objects if technologies will be designed to augment the surroundings.

This study consists of three phases: an initial interview, a contextual inquiry study on important objects in activities of daily living, and a co-design session for augmentations for important objects. The entire study will last no more than 2 hours. We will provide a compensation of 25 dollars per hour.

You will be audio and video recorded during your participation in this study. Audio and video recordings in their original form will only be accessible to the research team. Anonymized images or videos may be included in publications, academic presentations, and classrooms. The recordings will be retained indefinitely to be used in future research. Your personal information will remain confidential unless required by local law regarding mandatory reporting during home visits.

Now we will have your oral consent for the study. Would it be ok if we record the process?

1. **Do you have any questions about this research study?**
2. **Do you agree to participate in this study?**
3. **What is your name and what is the date for today?**
4. **Do you agree to be quoted directly in publications without your name?**

Pre-study interview

First, I am going to ask about your demographic information and visual condition.

1. How old are you?
2. How would you identify your gender?
3. What is your visual condition?
    a. Are you Legally blind?
    b. What is your Diagnosis?

    c. What is your Visual acuity?
    d. What is your Field of view?
    e. How is your contrast sensitivity?
    f. How is your Color vision?
    g. Are you sensitive to light?
4. How long have you had this visual condition?
    a. Is this condition progressive or stable?
5. Are you using any tools to assist with your visual experience?
6. This study is based on Augmented Reality, or AR. It is a technology that overlays digital information onto the real world to create an interactive experience between digital and physical elements. Do you have any prior experience with Augmented Reality (Google Glass, Hololens, phone applications)?
    a. If yes, could you share the experience?

Demonstrate the system prototype

First, I will show you an AR prototype. It's based on this AR headset called Hololens. On this device, there is a head-mounted display that you can see through, so you can see real-world objects and contents on the screen at the same time.

[Launch app before handing to participant]
Would you like to put it on? There is a knob at the back, you can spin it to adjust the tightness. There is a strap on the top, you can adjust it to fit your head. Make sure you can see through the display.

Are you seeing anything on the display?
[check on portal]
(if yes) would you hand me the headset? [close any pop-up windows] Thank you, now you can put the headset back on.

[Visualize a bottle]
Do you see a yellow outline around the bottle? When you move your head, the outline moves to always track the bottle.

This is the system we are building. It recognizes the bottle and augments it by outlining the bottle. Our goal is to augment objects in an environment with appropriate designs so that we can help you better perceive the surrounding environment. Therefore, we first want to understand what objects you want to augment with such a system.

(Definition: augment → add visual elements on the screen to make an object more visible, such as outline, icons, etc.)

Understand what objects to augment in kitchen environments

[contextual inquiry: kitchen]
Let's first stand at the entrance of the kitchen and look around. Imagine you come to your kitchen to start preparing a meal.

- What objects would you consider visually challenging and want to augment? Why?
- Beyond the challenging objects, what other objects do you think are important to augment to better perceive this kitchen environment? Why?

Now let's go closer to the kitchen. Imagine now you are preparing a meal in this kitchen.

- What objects would you consider visually challenging to see and want to augment during meal preparation? Why?
- Beyond the challenging objects, what other objects do you think are important to augment to better perceive this kitchen environment during meal preparation? Why?

Probe: knife, pans, pots, woks, stove, oven, spice containers, fridge, cutting board, measuring cups. Low contrast objects.

Probe one object:

- Did you see XXX?
    - (if not) Is this object visually challenging to you?
- Do you want to augment this object? Why?
- For other objects: "what about this?"

[watch video: kitchen]
Challenging actions: (1) navigate the kitchen, locate kitchen elements; (2) use stove and ovens, locate hot pans; (3) use sharp tools: cutting, chopping, peeling; (4) transferring, measuring; (5) finding dirty spots (hygiene);
We have discussed your kitchen environment. Now, let's turn to other kitchen environments that you are not familiar with. We are going to watch a video about different kitchen environments and a person performing different cooking activities. During the video, imagine yourself being the person doing the activities. Please think aloud, talking about what objects you think should be augmented in the video frames while you are watching the video. I will also pause the video at some frames to discuss certain objects.

Video segments: (1) kitchen overview; (2) see a lot of small objects; (3) search in the fridge; (4) cutting; (5) transferring; (6) using cooking tools; (7) shredding; (8) pouring; (9) measuring; (10) transferring; (11) turning food; (12) cleaning.

Probe:
00:05 (seg 1) → stove, sink, fridge, microwave, oven, dining table, trash can, trash
00:20 (seg 1) → toaster
00:39 (seg 2) → spice containers
00:50 (seg 2) → cutting board, scissors, bowls, plates, measuring cups
01:03 (seg 3) → containers
01:22 (seg 4) → cutting board, cucumber (food), knife
01:43 (seg 4) → stove, stove knob, pan
02:16 (seg 5) → pans, food, knife
02:24 (seg 6) → pans, spatula
02:35 (seg 6) → salt container, pans
03:03 (seg 7) → grater, carrot (food), hand
03:40 (seg 8) → kettle, pot, pan
04:00 (seg 9) → salt container, pot, spoon
04:27 (seg 10) → pan, bowl
04:43 (seg 10) → pan, plate
04:57 (seg 11) → tongs, pan
05:16 (seg 12) → sponge, bottle of detergent

[Summarize objects to augment]
We have discussed all the objects you want to augment in the kitchen environment. Let's discuss and compare these objects: which one you want to prioritize augmenting. [read the objects one by one] On a scale of 1 to 5, with 1 being the lowest priority and 5 being the highest priority, how would you rate the priority of augmenting this object? Why?

Thank you! That's all for this part of the study.

Understanding what objects to augment in outdoor environments

We have discussed objects you want to augment in the kitchen environment. Now, let's focus on the outdoor environment.

[Contextual Inquiry: walk]

First, let's take a walk outside. [show the specific route] The walk will take around 10 to 15 minutes. During the walk, please think aloud, talking about what objects you think should be augmented in the surroundings while you are navigating on the street. I will also discuss certain objects with you during the walk. Of course, safety first, we will not have any discussion when it's dangerous, say when crossing the street.

Probe: curb, traffic light, traffic lines, street signs, bollards (poles), bus signs, trash bins, manholes, construction facilities

Probe one object:

- Did you see XXX?
    - (if not) Is this object visually challenging to you?
- Do you want to augment this object? Why?
- For other objects: "what about this?"

[Contextual Inquiry: watch video]
We have discussed the outdoor environment in your neighborhood. Let's turn to other outdoor environments you may not be familiar with. We will watch a video about navigating on a busy street. Again, during the video, imagine yourself being the person navigating on the street. Please think aloud, talking about what objects you think should be augmented in the video frames while you are watching the video. I will also pause the video at some frames to discuss certain objects.

Probe:
00:08 → people in front; incoming car; metal fence on the left.
00:13 → trash can
00:15 → bus sign
00:38 → traffic light; zebra crossing line; incoming bike;
00:51 → storefront;
00:59 → traffic sign pole; construction site;
01:25 → Obstacle on sidewalk; curb;
01:34 → tree;
02:10 → double confirm: traffic sign, pole, tree

[Summarize objects to augment]
Here is the list of all the objects you want to augment in the kitchen environment. Let's discuss and compare these objects, and talk about which objects you want to prioritize when augmenting them. [read the objects one by one] On a scale of 1 to 5, with 1 being

the lowest priority and 5 being the highest priority, how would you rate the priority of augmenting this object? Why?

Understanding the characteristics of visually important and challenging objects in different scenarios

So far, we have discussed objects you want to augment in the kitchen and outdoor environment, including objects that are challenging to perceive and objects that are important to augment regardless of the visibility. Let's expand to other types of activities of daily living, such as buying groceries, taking a bus, or eating in a restaurant. Try to think of these different scenarios and the objects you want to augment in these scenarios.

- What characteristics make an object visually challenging to you so that you want to augment?
- What characteristics make an object important to be augmented regardless of visibility?
    - Try: What characteristics makes an object the one you want to augment even if it's not visually challenging?

Evaluate and brainstorm different designs for important object augmentation

Now let's talk about how you would like to augment these objects.
First, we will discuss your preferred design to augment objects. We will go through several design examples and discuss each of them. Be aware that these are only examples. After going through them, we will also ask you to brainstorm new design ideas that you would prefer to use, and it does not have to be visual feedback.

(solid) Here's a design where we put a solid overlay on each object. Let's look at this design in different scenarios: two kitchens, two outdoor.

- Do you think this design would be effective in augmenting the objects? Why?
- What scenarios do you think this design would be suitable for?
- What scenarios do you think this design would not be suitable for?
- How do you want to improve the design?

(static outline) Here's a design where we outline each object. Let's look at this design in different scenarios: two kitchens, two outdoor.

- Do you think this design would be effective in augmenting the objects? Why?
- What scenarios do you think this design would be suitable for?

- What scenarios do you think this design would not be suitable for?
- How do you want to improve the design?

(flashing outline) Here's a design where we outline each object and the outline is flashing. Let's look at this design in different scenarios: two kitchens, two outdoor.

- Do you think this design would be effective in augmenting the objects? Why?
- What scenarios do you think this design would be suitable for?
- What scenarios do you think this design would not be suitable for?
- How do you want to improve the design?

(text) Here's a design where we put a text label on top of each object with its type. Let's look at this design in different scenarios: two kitchens, two outdoor.

- Do you think this design would be effective in augmenting the objects? Why?
- What scenarios do you think this design would be suitable for?
- What scenarios do you think this design would not be suitable for?
- How do you want to improve the design?

Preference: We have seen four designs: solid overlay, static outline, flashing outline, and text label. [click button on the last page to review all examples]

- Which one would you prefer most? Why?

Brainstorm: Now let's brainstorm!
What are some other design ideas you have in mind to augment objects? Besides the visual designs we have seen, you can also think of audio and haptic feedback.

- Can you describe the design? (We have papers and pens if you want to draw)
- Why do you think it would be effective?
- What scenarios do you think this design would be suitable for? Why?
- What scenarios do you think this design would not be suitable for?

We have seen different augmentation designs, but so far, we assumed all objects are equally important and we augmented them in the same way.

- Among these objects, do you think they should be augmented equally? Why?
- Do you want to use different designs to distinguish objects with different priorities? Why?

Now, let's talk about how you would like to use different designs to distinguish objects with different priorities. We will go through several examples and discuss each of them. Be aware that these are only examples. After going through them, we will also ask you

to brainstorm new methods that you would prefer to distinguish objects with different priorities.

(form) Here's a method where we use different forms of augmentation to distinguish objects with different priorities. We use a solid overlay for some objects and outline for others. Let's look at this method in different scenarios: two kitchens, two outdoor.

- Do you think this method would be effective in differentiating objects? Why?
- What scenarios do you think this method would be suitable for?
- What scenarios do you think this method would not be suitable for?
- How do you want to improve the method?

(color) Here's a method where we use different colors to distinguish objects with different priorities. We use yellow for some objects and orange for others. Let's look at this method in different scenarios: two kitchens, two outdoor.

- Do you think this method would be effective in differentiating objects? Why?
- What scenarios do you think this method would be suitable for?
- What scenarios do you think this method would not be suitable for?
- How do you want to improve the method?

(text) Here's a method where we use text labels to distinguish objects with different priorities. We use dark-yellow outlines for all objects, and add text labels to some objects but not the others. Let's look at this method in different scenarios: two kitchens, two outdoor.

- Do you think this method would be effective in differentiating objects? Why?
- What scenarios do you think this method would be suitable for?
- What scenarios do you think this method would not be suitable for?
- How do you want to improve the method?

(visual effect) Here's a method where we use different visual effects to distinguish objects with different priorities. We use flashing outlines for some objects and static outlines for others. Let's look at this method in different scenarios: two kitchens, two outdoor.

- Do you think this method would be effective in differentiating objects? Why?
- What scenarios do you think this method would be suitable for?
- What scenarios do you think this method would not be suitable for?
- How do you want to improve the method?

Preference: We have seen four methods of distinguishing objects with different priorities: different forms, different colors, text labels, and different visual effects. [click button on the last page to review all examples]

- Which one would you prefer? Why?

Brainstorm: Now let's brainstorm!
What are some other methods you have in mind that distinguish objects with different priorities? Besides the visual designs we have seen, you can also think of audio and haptic feedback.

- Can you describe how you would distinguish the objects?
- Why do you think it would be effective?
- What scenarios do you think this method would be suitable for? Why?
- What scenarios do you think this method would not be suitable for?

Ending

Thank you for your answers. Your response is very valuable for our study. Would you like to be contacted for future research on low-vision technologies and other relevant topics?
Thank you. That is the end of the study.
I will send you the consent form and your oral consent shortly. Would you prefer using email or text?